

TandemSDR v1.0

TandemSDR is a self-contained SDR receiver front-end that runs entirely in a browser with no installation required. It connects to SDRplay's SDRconnect software via WebSocket and provides a full receiver control interface including spectrum display, waterfall, audio output, signal metering (dBm and SNR), and signal history charting.

Features

- **WebSocket connection to SDRconnect**
 - Connects to SDRconnect running locally or on a network.
 - Default connection: 127.0.0.1 port 5454 (both configurable).
 - Will connect over the WAN if router has port 5454 forwarding enabled.
 - Connection status indicator shows Connected, Connecting, or Disconnected.
 - On connect, all current device state is retrieved automatically from SDRconnect.
- **Frequency Control**
 - Separate VFO and Center frequency inputs in MHz.
 - Hold-to-repeat step buttons and mouse scroll wheel for VFO tuning. Step buttons accelerate the longer they are held.
 - Selectable step size: 10 Hz, 100 Hz, 500 Hz, 1 kHz, 2.5 kHz, 5 kHz, 10 kHz, 12.5 kHz, 25 kHz, 100 kHz.
 - Hold-to-repeat step buttons for Center frequency using the selected step size.
 - VFO → Center button to sync the center to the current VFO frequency. When sample rate is above 2 MSPS a 10 kHz offset is applied to keep the VFO out of the DC centre of the spectrum.
- **Device Controls**
 - Sample rate selector: 62.5 kSPS to 10 MSPS.
 - Device selector — populated dynamically from SDRconnect.
 - Antenna selector — populated dynamically from SDRconnect.
 - RF Gain slider with overload warning indicator.
- **Demodulation**
 - Modes: AM, SAM (Synchronous AM), USB, LSB, CW, NFM, WFM.
 - Each mode sets a default filter bandwidth on selection.

- Filter bandwidth input with hold-to-repeat step buttons that accelerate the longer they are held. Maximum bandwidth is enforced per mode.
- WFM mode requires a minimum sample rate of 250 kSPS; the 62.5 and 125 kSPS options are disabled when WFM is selected. If either of those rates is active when WFM is selected, the sample rate automatically jumps to 2 MSPS.
- RDS on/off toggle with PI, PTY, and PS display (WFM mode).
- Stereo on/off toggle with stereo indicator.
- **Audio**
 - Audio streamed from SDRconnect and decoded in the browser via the Web Audio API.
 - Volume control slider with mute toggle button.
 - Noise Reduction on/off toggle with strength slider.
 - Audio AGC on/off toggle with threshold slider.
 - Squelch on/off toggle with level slider.
- **S-Meter**
 - Segmented bar display calibrated to IARU standard.
 - HF mode (below 30 MHz): S9 = -73 dBm, 6 dB per S-unit.
 - VHF mode (30 MHz and above): S9 = -93 dBm, 6 dB per S-unit.
 - Segments S1–S9 in green, +10 through +60 dB over S9 in red.
 - Automatically switches HF/VHF calibration based on current frequency.
- **Signal readouts**
 - Live Signal Power display in dBm.
 - Live Signal SNR display in dB.
 - Live Frequency display.
- **Spectrum Display**
 - FFT spectrum analyzer. (512 points)
 - Adjustable Max (reference) and Min (base) level inputs.
 - Frame averaging: None, 2, 4, 8, 16 frames.
 - Zoom: ×1, ×2, ×4 — cycled via button or keyboard.
 - Frequency snap grid: 1 kHz, 5 kHz, 9 kHz, 10 kHz, 25 kHz, 100 kHz.
 - Crosshair follows mouse with frequency readout.

- Left-click sets VFO frequency at cursor position.
- Right-click sets Center frequency at cursor and syncs VFO.
- Drag on the frequency label row to pan the center frequency.
- Scroll wheel steps the VFO frequency.
- **Waterfall Display**
 - Scrolling waterfall below the spectrum.
 - Optional frame averaging (follows spectrum averaging setting).
 - Speed selector: Slow, Medium, Fast.
 - Same mouse controls as the spectrum (left-click, right-click, scroll wheel).
- **Signal History Chart**
 - Time-series chart of Signal Power (dBm) or SNR over time.
 - Selectable capture interval: 500 ms, 1 sec, 5 sec, 10 sec, 30 sec, 60 sec.
 - Start/Stop/Clear capture control.
 - Session min and max values with timestamps displayed on the chart.
 - Scrollable history slider to review up to 10× the visible window of captured data. Automatically snaps back to live after 20 seconds of inactivity.
 - Crosshair mouseover shows exact value and timestamp when stopped or in history mode.
 - **Record** — opens a native file save dialog and streams data directly to a CSV file in real time as samples are captured.
 - **Export** — saves the current in-memory buffer to a CSV file via a native file save dialog. CSV includes frequency, mode, capture interval, and timestamped power and SNR columns.
- **Memory**
 - 10 memory slots for saving and recalling receiver settings.
 - Each slot stores VFO frequency, center frequency, mode, filter bandwidth, sample rate, antenna, RF gain, step size, volume, and spectrum display settings (reference level, base level, zoom, averaging, snap).
 - Slots are saved in browser local storage and survive page reloads.
 - Individual slots can be cleared from the memory modal.
- **Band Selector**
 - Quick-access modal for jumping to common bands with one click.

- Sets center frequency, sample rate, mode, snap, and step for the selected band. Also sets RF gain to maximum and turns squelch off.
- Covers Amateur Radio (160 m through 33 cm), Shortwave Broadcast (120 m through 11 m), and Broadcast (AM and FM).

- **Keyboard Shortcuts**

Key	Action
← →	Step VFO by selected step size
↑ ↓	Step Center by Snap value
PgUp PgDn	Jump Center one full span width
=	Copy VFO → Center
A	AM mode
S	SAM mode
U	USB mode
L	LSB mode
C	CW mode
N	NFM mode
W	WFM mode
V	Toggle audio mute
Q	Toggle squelch
D	Toggle Spectrum <u>D</u> isplay
F	Toggle Water <u>F</u> all (enables Spectrum first if hidden)
Z	Cycle spectrum zoom (×1 → ×2 → ×4)
+	Increase sample rate
-	Decrease sample rate
M	Save memory slot
R	Recall memory slot
B	Open Band Selector
H or ?	Open Help
Esc	Close any open modal

- **Built-in Help**

- Help modal accessible via the ? button or H/? keyboard shortcut.
- Covers all mouse controls for spectrum and waterfall.

- Full keyboard shortcut reference.

Requirements

- No installation and no internet connection required. Open the HTML file directly in your browser
- **SDRplay SDRconnect** must be running with WebSocket access enabled on port 5454 (default).
- Compatible with any SDRplay RSP device supported by SDRconnect.
- Works in any modern browser. The **Record** and **Export** features require a Chromium-based browser (Google Chrome, Microsoft Edge, Brave, or Opera) as they depend on the File System Access API, which is not supported in Firefox or Safari.

Typical Workflow

1. Start **SDRconnect** and ensure it is running with WebSocket enabled.
2. Open **TandemSDR** by double-clicking the HTML file in your browser.
3. Enter the IP address and port of the SDRconnect instance (defaults to 127.0.0.1:5454).
4. Click **Connect**. TandemSDR will retrieve all current device settings automatically.
5. Use the **VFO** input or click on the **Spectrum** to tune to a frequency.
6. Select a demodulation **Mode** and adjust **Filter BW** as needed.
7. Adjust **RF Gain**, **Volume**, **Noise Reduction**, and **Squelch** to taste.
8. Click **Show** under Spectrum Display to enable the spectrum and waterfall.
9. Use **Signal History** to capture and export signal power or SNR over time.

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